Amino acid metabolism

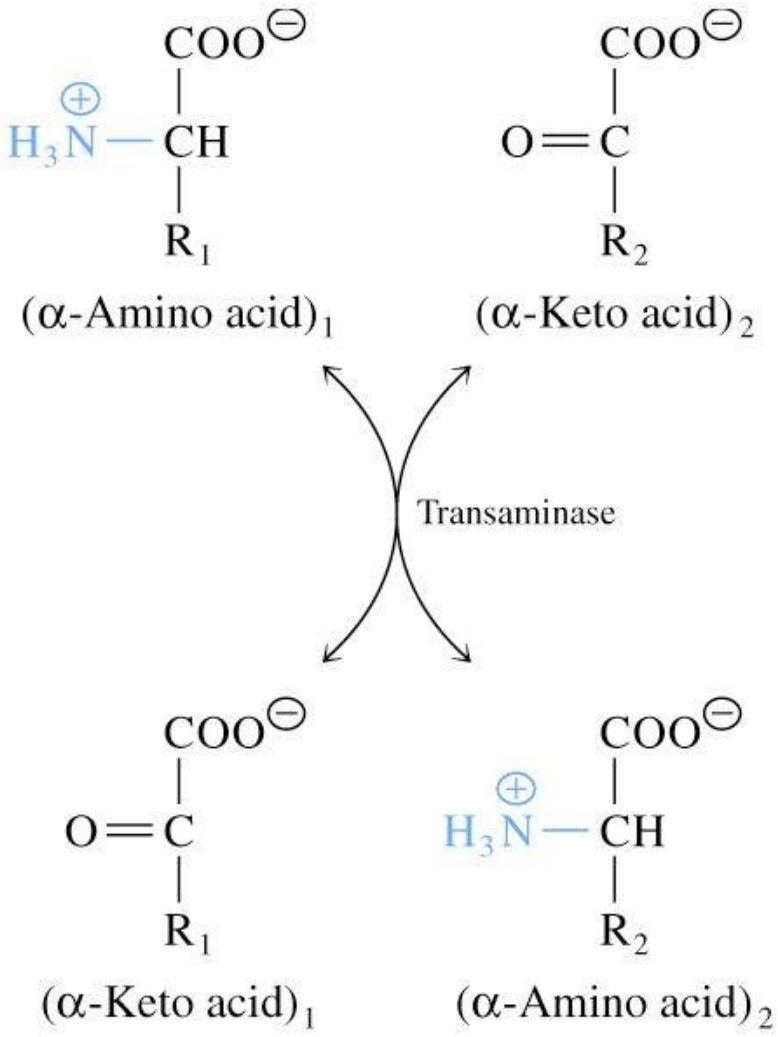
- Metabolism of amino acids differs, but 3 common reactions:
  - Transamination
  - Deamination
  - Decarboxylation

Transamination

In transamination

- Amino acids are degraded in the liver.
- An amino group is transferred from an amino acid to an α-keto acid, usually α-ketoglutarate.
- The reaction is catalyzed by a *transaminase or aminotransferase.*
- A new amino acid, usually glutamate, and a new α-keto acid are formed.

Transamination reactions



Enzymatic Transamination

- Typically, α-ketoglutarate accepts amino groups
- L-Glutamine acts as a temporary storage of nitrogen
- L-Glutamine can donate the amino group when needed for amino acid biosynthesis
- All aminotransferases rely on the pyridoxal phosphate cofactor

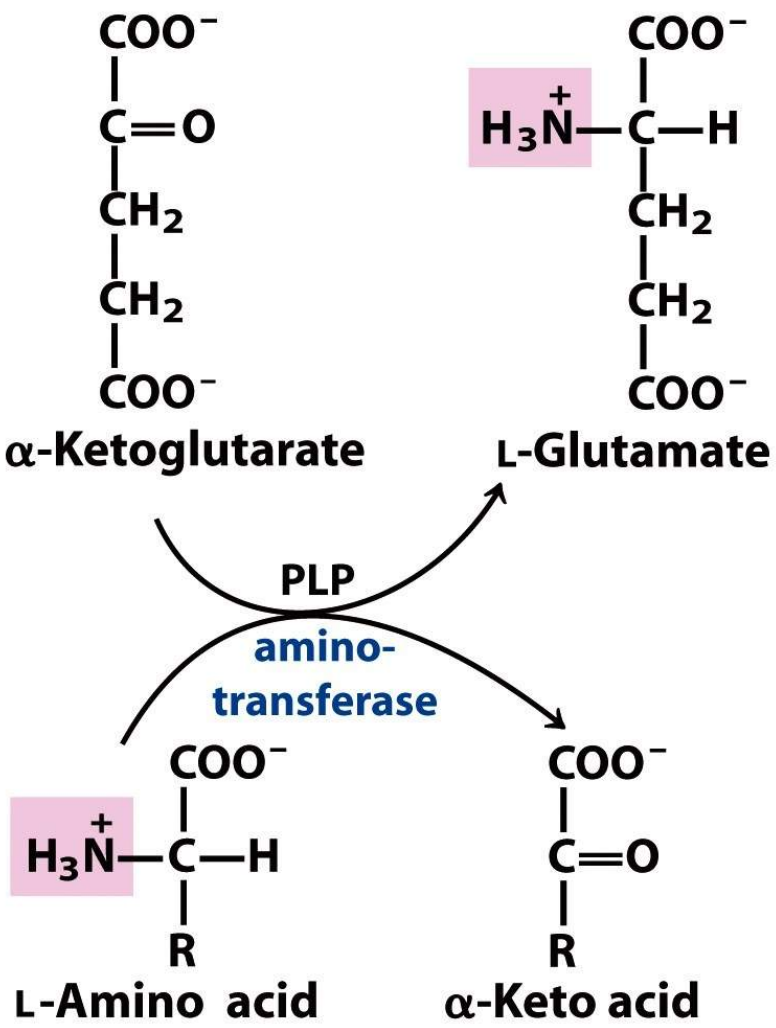
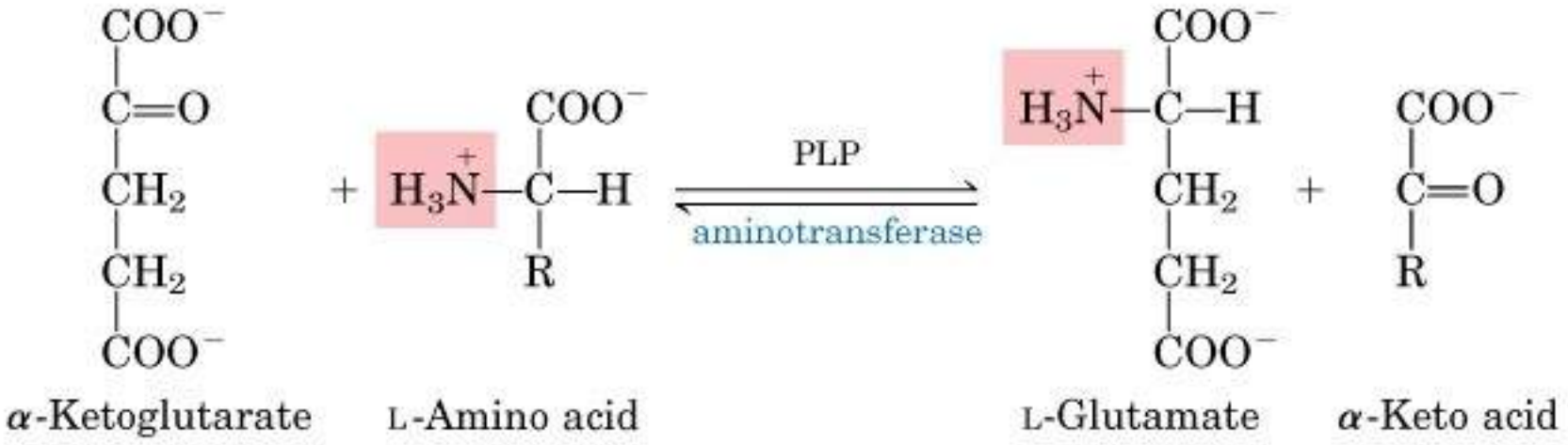


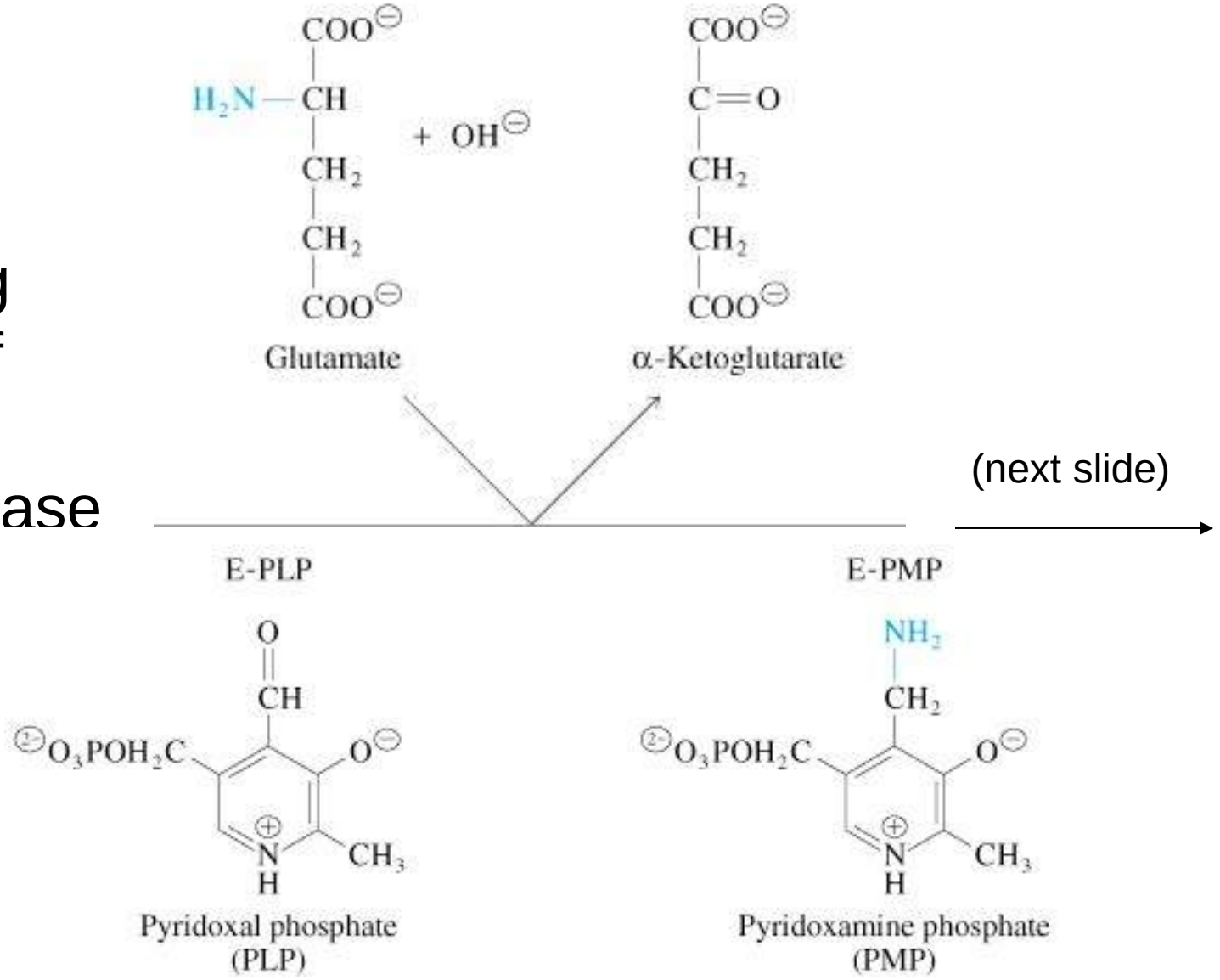
Figure 18-4
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Amino Group Transfer - Aminotransferase

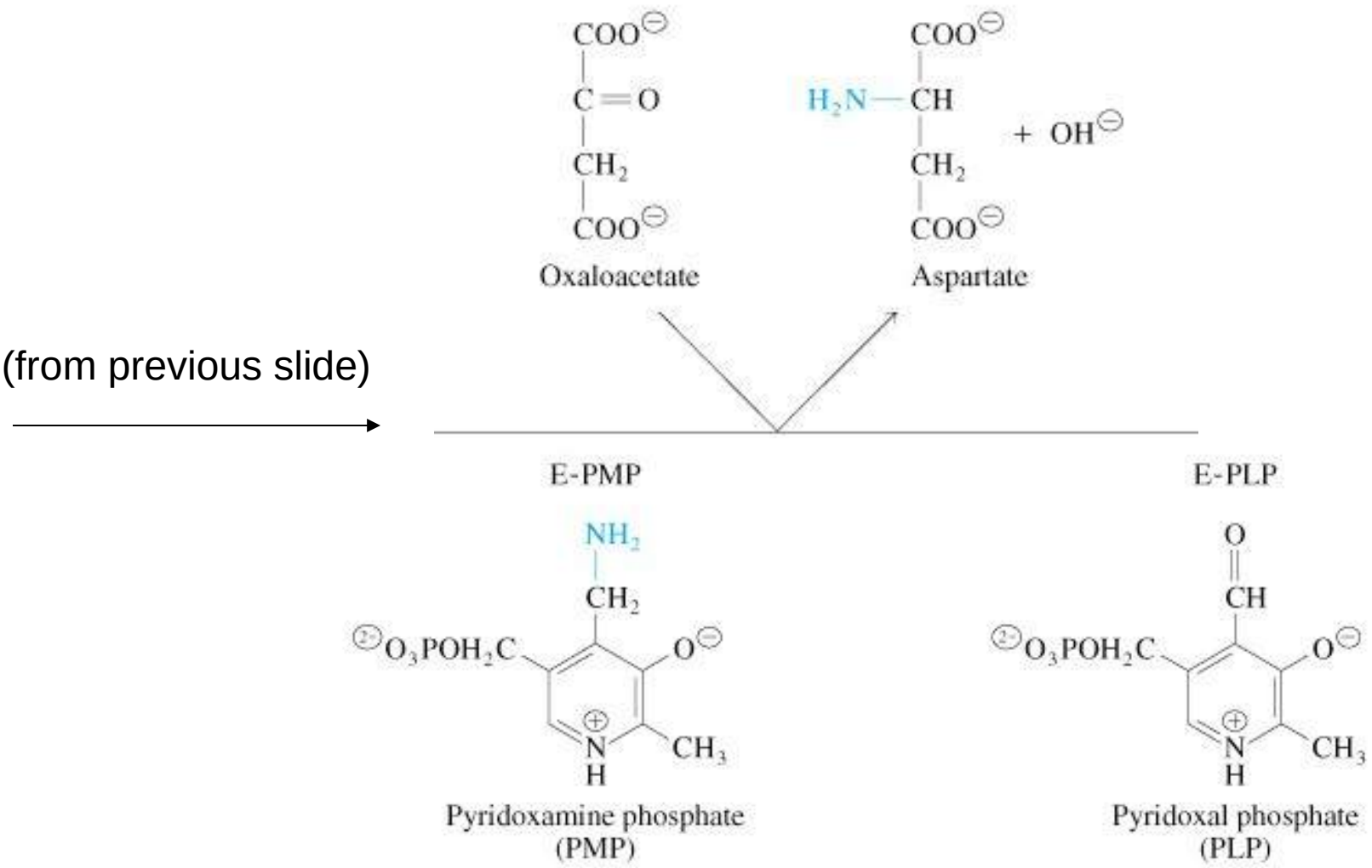
Enzymatic removal of α-amino groups (transaminase /aminotransferases - named for amino donor
i.e. Ala aminotranferase removes amino group from Ala)



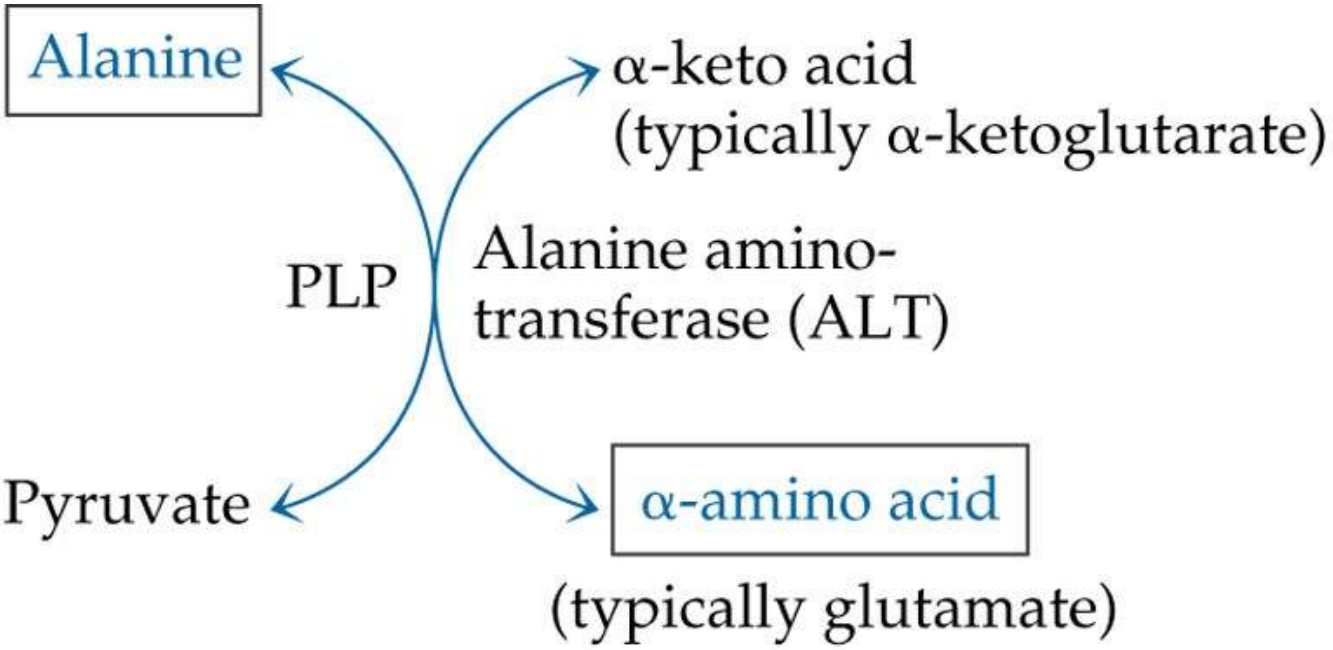
- Ping-pong kinetics of aspartate transaminase

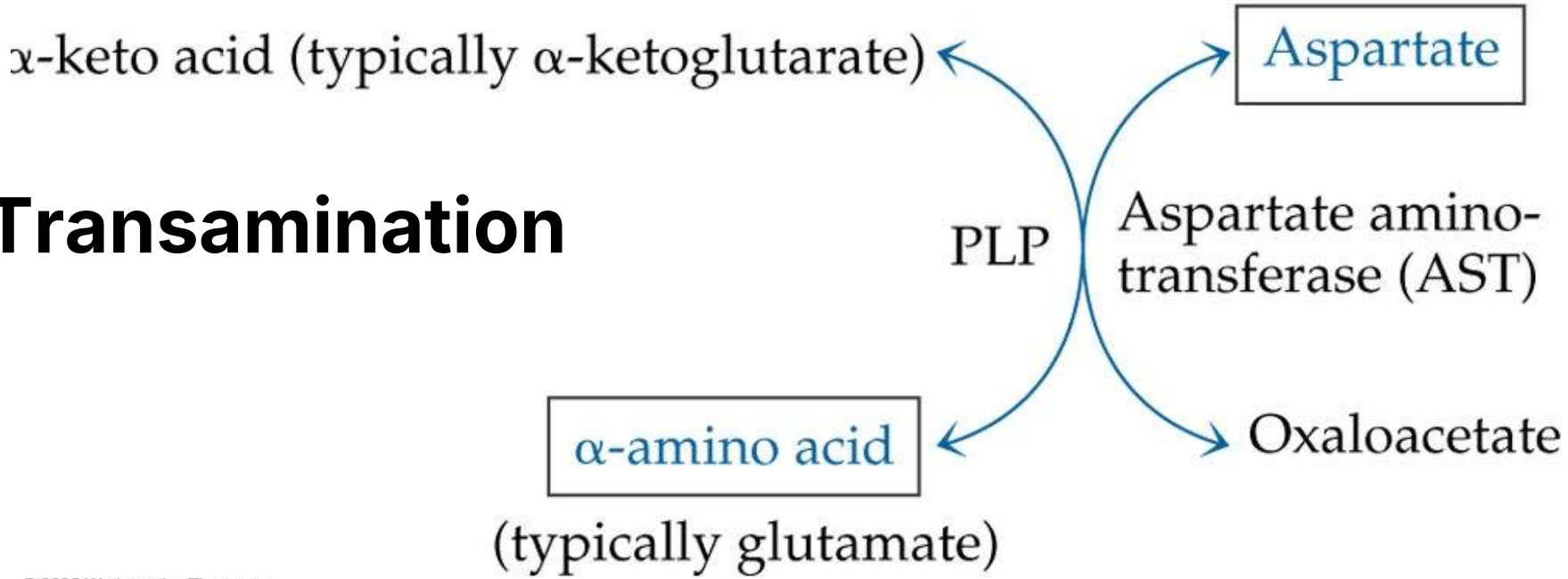


(from previous slide)

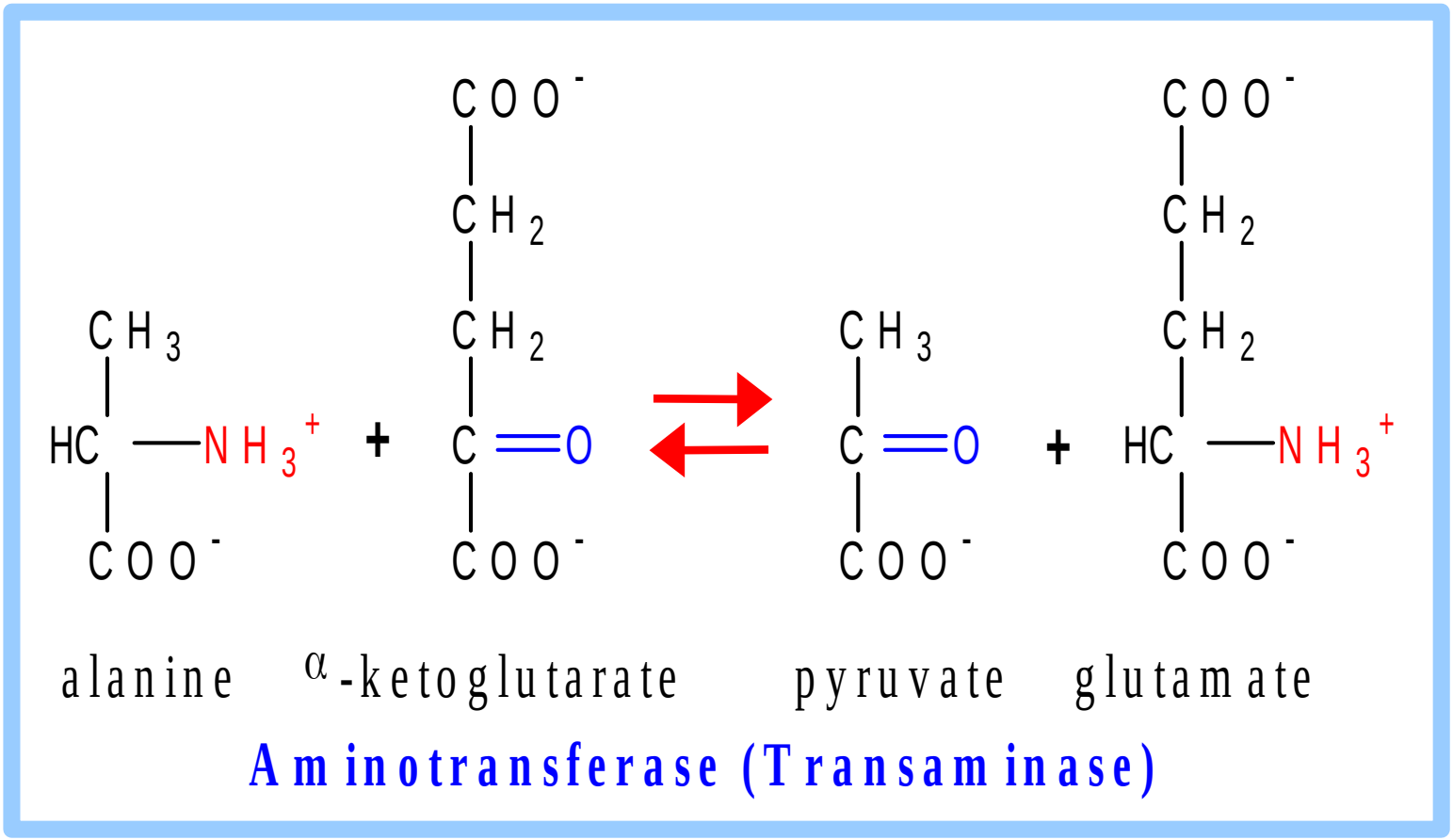


Transamination



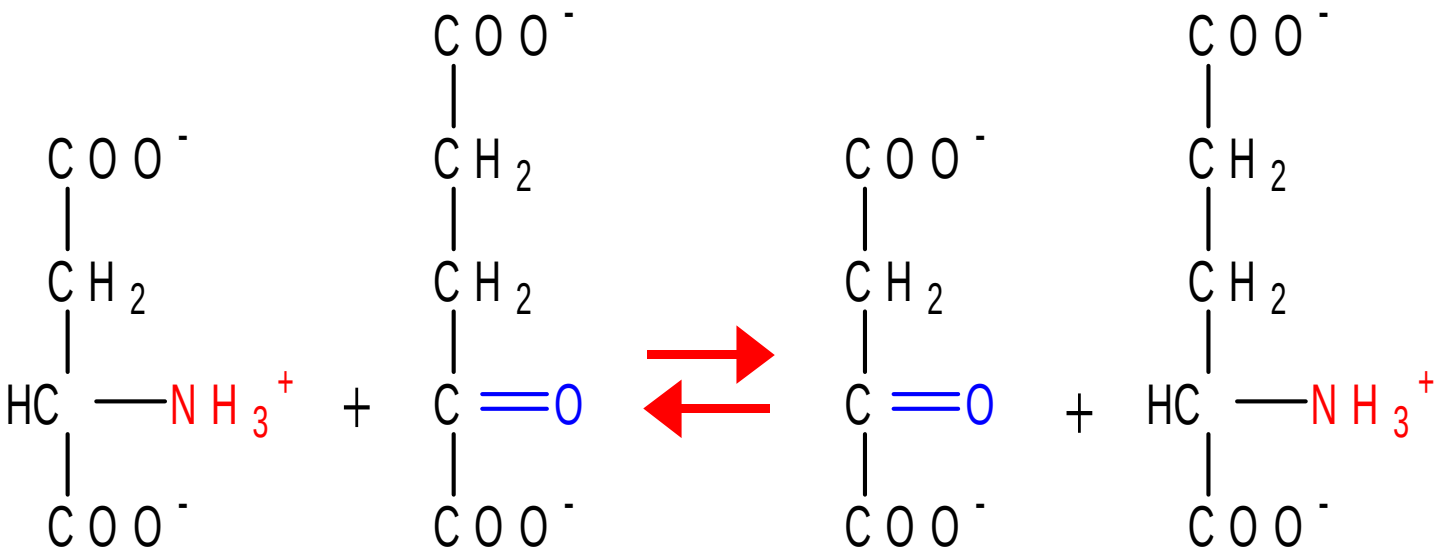


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The **3-C** α-keto acid **pyruvate** is produced from **alanine, cysteine, glycine, serine, & threonine**.

Alanine deamination via Transaminase directly yields pyruvate.



The **4-C** Krebs Cycle intermediate **oxaloacetate** is produced from **aspartate & asparagine**.

Aspartate transamination yields **oxaloacetate**.

Aspartate is also converted to **fumarate** in Urea Cycle. Fumarate is converted to oxaloacetate in Krebs cycle.

The Amino Group is Removed From All Amino Acids First

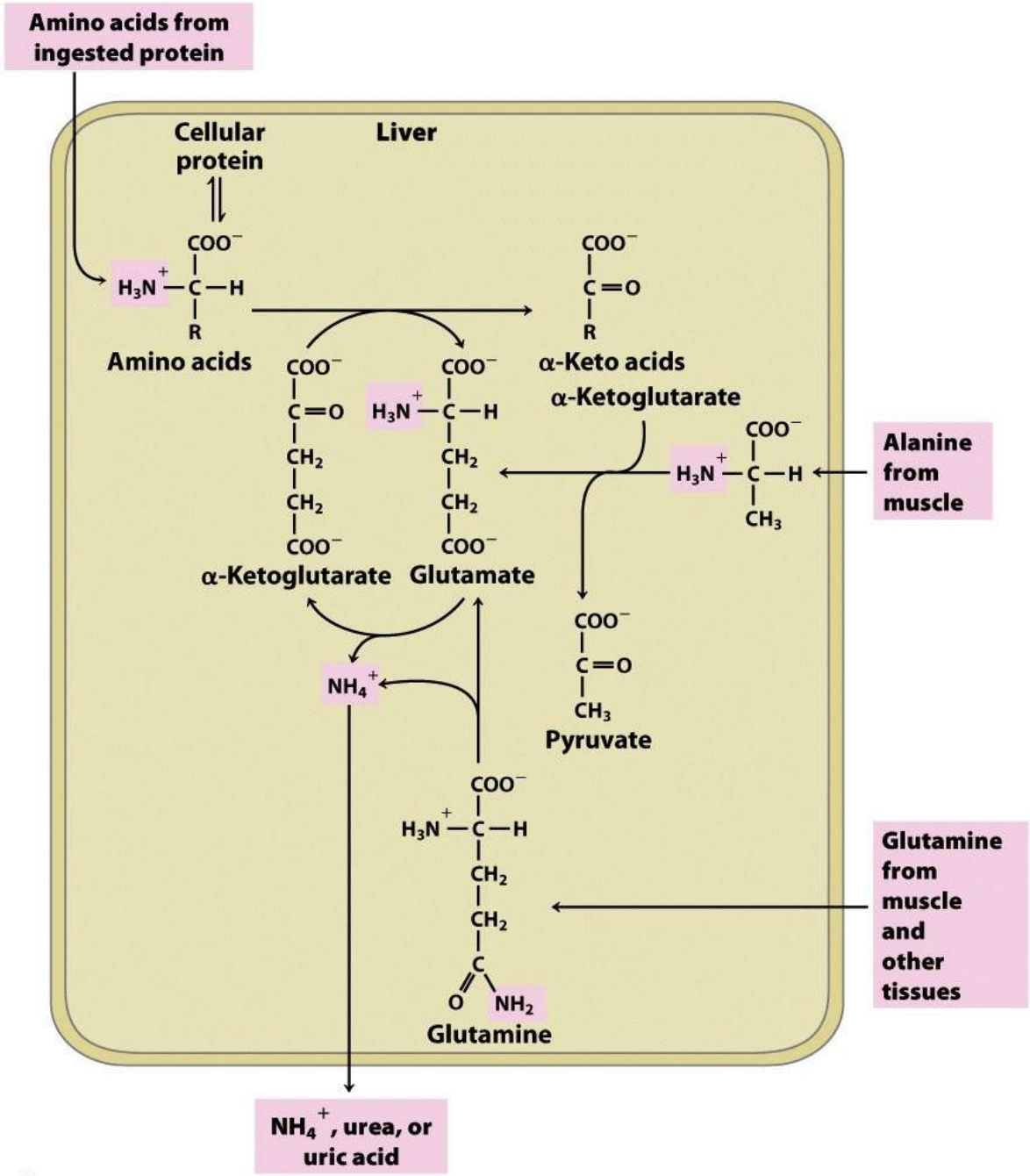


Figure 18-2a

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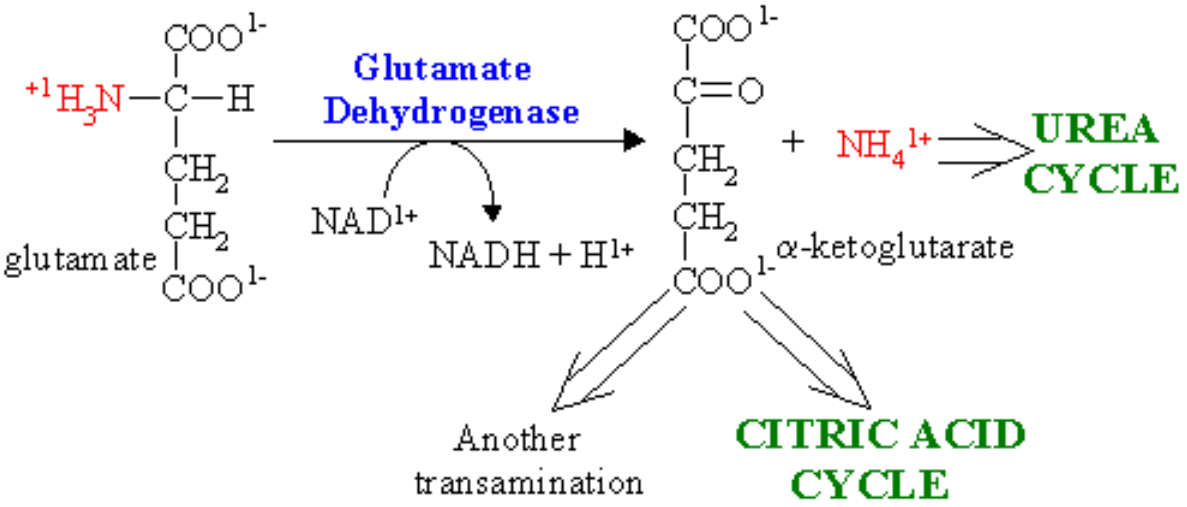
Oxidative Deamination

Oxidative deamination

- Removes the amino group as an ammonium ion from glutamate.
- Provides α-ketoglutarate for transamination.

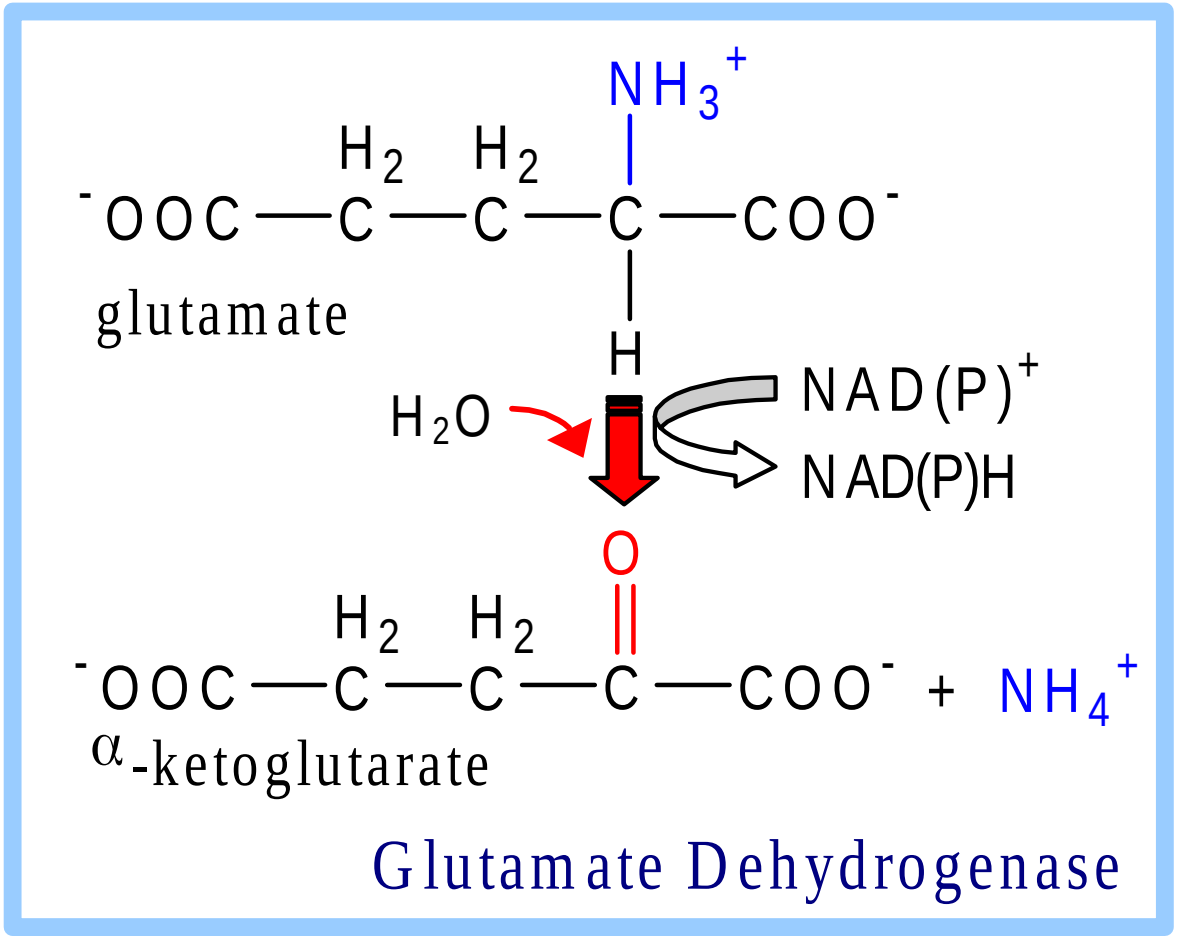
Oxidative Deamination

- Glutamate formed by transamination reactions is deaminated to α-ketoglutarate
- Glutamate dehydrogenase - NAD^+ or $NADP^+$ is coenzyme



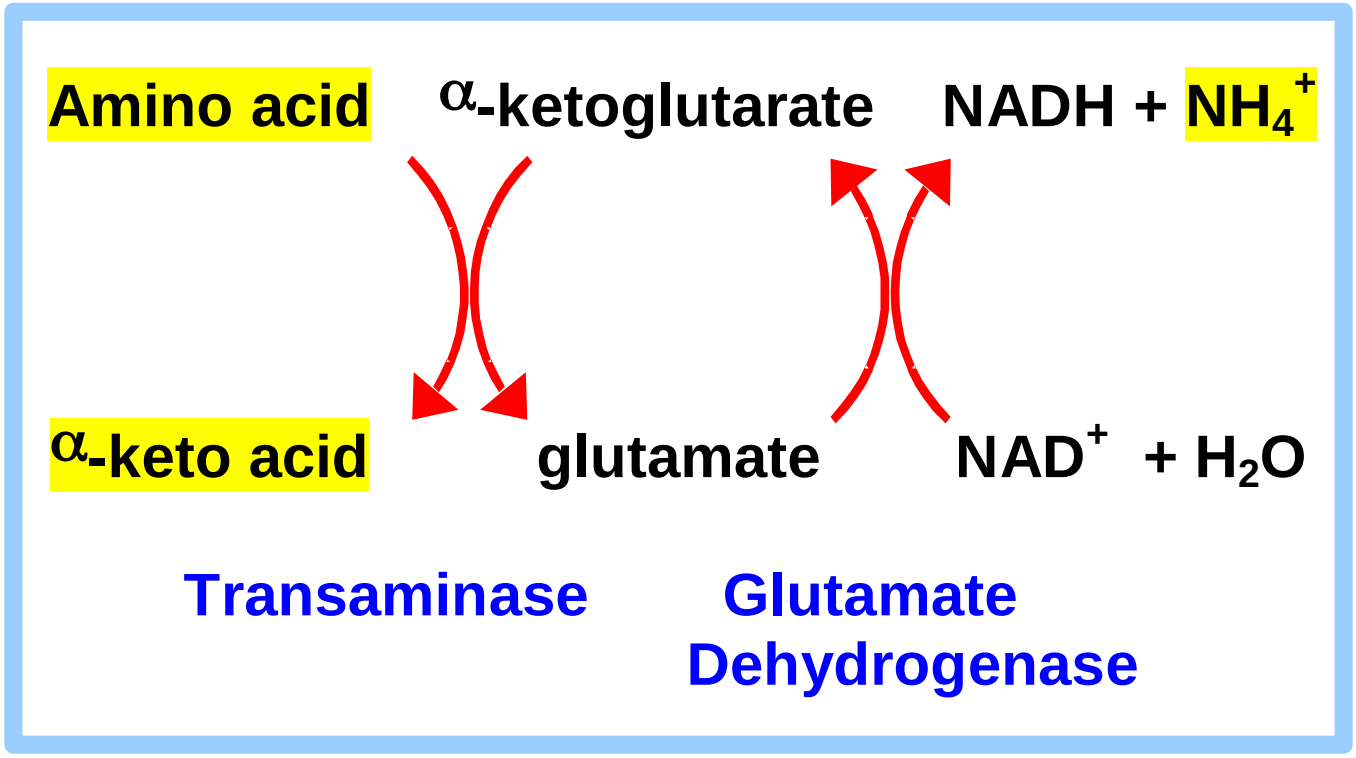
- Other AA oxidases - (liver, kidney) low activity

Glutamate Dehydrogenase catalyzes a major reaction that effects **net removal of N** from the amino acid pool.



It is one of the few enzymes that can use **NAD^+ or $NADP^+$** as e^- acceptor.

Oxidation at the α-carbon is followed by hydrolysis, releasing NH_4^+.



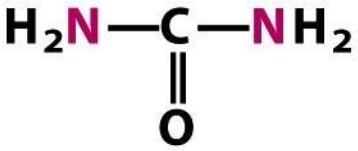
Summarized above:

The role of transaminases in funneling amino N to glutamate, which is deaminated via Glutamate Dehydrogenase, producing NH_4^+.

Excretory Forms of Nitrogen



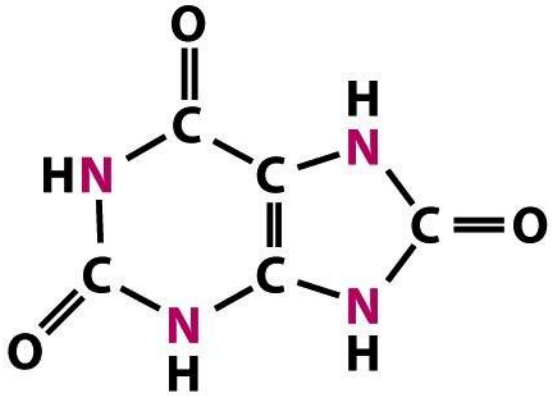
Ammonia (as ammonium ion)



Urea

Ammonotelic animals: most aquatic vertebrates, such as bony fishes and the larvae of amphibia

Ureotelic animals: many terrestrial vertebrates; also sharks



Uric acid

Uricotelic animals: birds, reptiles

Figure 18-2b
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Fate of Individual Amino Acids

- Seven to acetyl-CoA
  - Leu, Ile, Thr, Lys, Phe, Tyr, Trp
- Six to pyruvate
  - Ala, Cys, Gly, Ser, Thr, Trp
- Five to α-ketoglutarate
  - Arg, Glu, Gln, His, Pro
- Four to succinyl-CoA
  - Ile, Met, Thr, Val
- Two to fumarate
  - Phe, Tyr
- Two to oxaloacetate
  - Asp, Asn

Summary of Amino Acid Catabolism

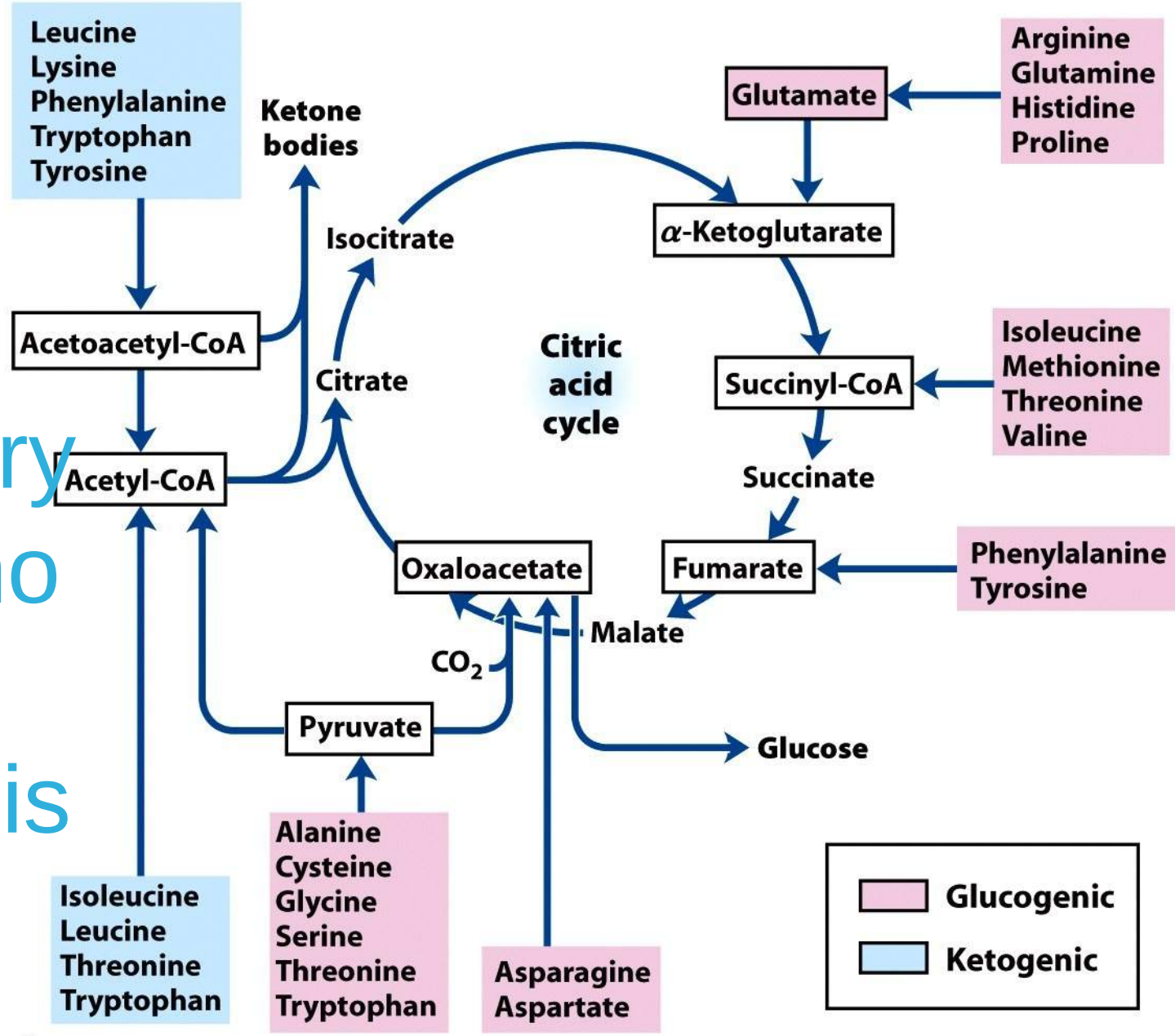


Figure 18-15
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